Supplementary Material for "Trust, but Verify: Online-Arbitrated Domain Knowledge for Active Data Acquisition from Expensive Black-Box Sources with Mixed Outputs"

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I. Data Sources

This section provides detailed descriptions of the six expensive data sources used in the experimental evaluation. For each source, we summarize the number of factors, response types, number of responses, initial sample size, and total sample size.

A. Branin

The classical Branin function is a two-dimensional test function widely used in global optimization. In our mixed-response setting, the input domain is scaled to $[0,1]^2$, and the original output is transformed to produce two continuous responses and one binary categorical response. Let $\mathbf{x} = [x_1, x_2]^\top$ with $x_1, x_2 \in [0,1]$. The scaled variables are

$$\tilde{x}_1 = 15x_1 - 5, \qquad \tilde{x}_2 = 15x_2. \tag{1}$$

The original Branin response is

$$y_{\mathrm{br}} = \Big(\tilde{x}_2 - \tfrac{5.1}{4\pi^2}\tilde{x}_1^2 + \tfrac{5}{\pi}\tilde{x}_1 - 6\Big)^2 + 10\big(1 - \tfrac{1}{8\pi}\big)\cos(\tilde{x}_1) + 10 + \tfrac{\tilde{x}_1}{3}. \tag{2}$$

The three outputs of the modified Branin function are then defined as

$$y_1 = 2\log(y_{\mathrm{br}}) - 3, \qquad y_2 = \big(\log(y_{\mathrm{br}})\big)^2, \tag{3}$$

$$y_3 = \mathbf{1}\big[\,\mathrm{sigmoid}(4 - \log y_{\mathrm{br}}) > 0.5\big]. \tag{4}$$

Here y_1 and y_2 are continuous responses, and y_3 is a binary categorical response. The experiments start with an initial training set of 30 samples, followed by 20 adaptively selected samples, for a total budget of 50 samples. Each step selects the next query from a candidate pool of 100 points.

B. Ackley

The Ackley function is a well-known multimodal benchmark with numerous local minima. We employ a six-dimensional variant that generates two continuous responses and one categorical response. Let $\mathbf{x} \in [0,1]^6$ and denote the scaled variable $\mathbf{z} = 2\mathbf{x}$. Denoting $\bar{r} = \sqrt{\frac{1}{6}\sum_{j=1}^{6} z_j^2}$, $\bar{c} = \frac{1}{6}\sum_{j=1}^{6}\cos(2\pi z_j)$, and $\bar{s} = \frac{1}{6}\sum_{j=1}^{6} z_j$, the two continuous responses are

$$y_1 = 20 + e - 20\,e^{-0.2\bar{r}} - e^{\bar{c}}, \qquad y_2 = 20 + e^{\bar{s}} - 20\,e^{-0.2\bar{r}} - e^{\bar{c}}. \tag{5}$$

Compared with y_1, y_2 introduces an exponential perturbation term $e^{\bar{s}}$ in the first additive component. The categorical response is obtained by thresholding a nonlinear combination of y_1 and y_2:

$$y_3 = \mathbf{1}[y_1 y_2 + y_1 + y_2 \geq 35]. \tag{6}$$

The experiments start with an initial training set of 20 samples, followed by 30 adaptively selected samples, for a total budget of 50 samples. Each step selects the next query from a candidate pool of 200 points.

C. Wingweight

The wing weight design problem is a classic aerospace engineering benchmark for estimating the weight of an aircraft wing from ten design parameters. The ten input factors are: wing area S_w, fuel weight W_{fw}, aspect ratio A, quarter-chord sweep angle Λ, dynamic pressure q, taper ratio λ, thickness-to-chord ratio t/c, ultimate load factor N_z, flight design gross weight W_{dg}, and paint weight W_p. All inputs are normalized to $[0,1]$.

With $\kappa = (100\,t/c)/\cos\Lambda$, $P_w = S_w^{0.758} W_{fw}^{0.003} \big(\frac{A}{\cos^2\Lambda}\big)^{0.6}$, and $P'_w = S_w^{0.6} W_{fw}^{0.0035} \big(\frac{A}{\cos^2\Lambda}\big)^{0.5}$, the two continuous wing weight responses are:

$$y_1 = 0.036\,P_w\,q^{0.006}\lambda^{0.04}\,\kappa^{-0.3}(N_z W_{dg})^{0.49} + S_w W_p, \tag{7}$$

$$y_2 = 0.038\,P'_w q^{0.006}\lambda^{0.05}\,\kappa^{-0.5}(N_z W_{dg})^{0.5} - S_w W_p. \tag{8}$$

The two continuous responses simulate two distinct wing configurations (the main wing and the auxiliary wing) on the same aircraft. The categorical response indicates whether the design satisfies the weight constraints $y_1 \leq 250$ and $y_2 \leq 50$:

$$y_3 = \mathbf{1}[y_1 \leq 250 \ \wedge \ y_2 \leq 50]. \tag{9}$$

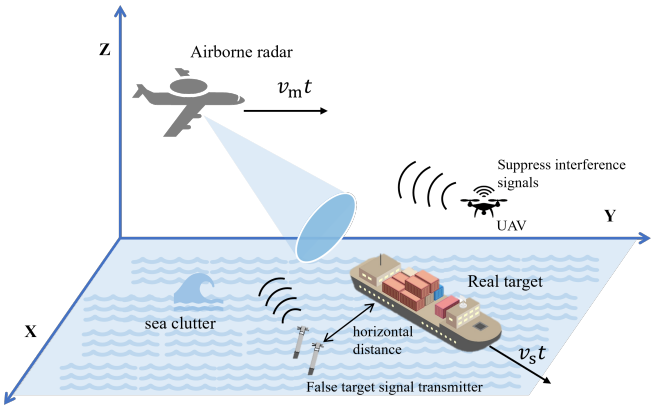


Fig. 1. Radar anti-jamming performance experiment scenario.

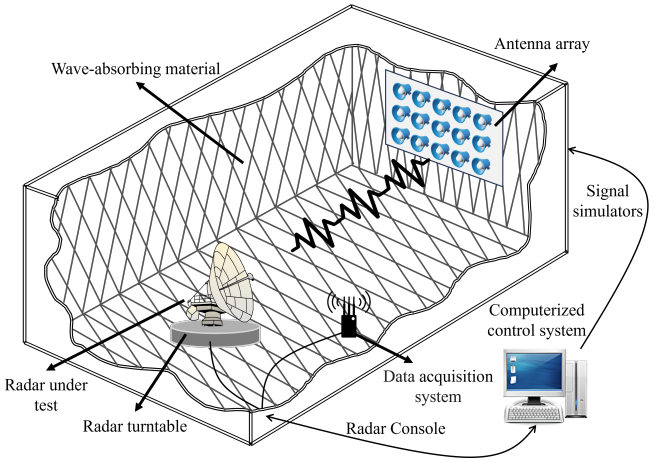


Fig. 2. Microwave anechoic chamber.

TABLE I
INFORMATION OF INPUT VARIABLES FOR THE SHIP CASE.

Interference Type	Factor Name	Range	Unit
Noise Suppression	Signal-to-noise ratio	40–70	dB
	Occupancy ratio	10–70	–
	End moment	20–45	s
	Start moment	10–35	s
Fake Target Deception	Signal-to-noise ratio	10–25	dB
	Fake targets dist.	30–400	m
	Horizontal distance	0.2–3.4	km

TABLE II
INFORMATION OF THE RESPONSES FOR THE SHIP CASE.

Symbol	Response	Category
y_1	Tracking time ratio of fake target	Continuous
y_2	SNR Binned Entropy	Continuous
z	Anti-jamming result	Categorical

The experiments start with an initial training set of 25 samples, followed by 20 adaptively selected samples, for a total budget of 45 samples. Each step selects the next query from a candidate pool of 200 points.

D. Ship

Finally, we consider a radar anti-jamming performance experiment. Such experiments evaluate the performance and reliability of radar systems in applications including maritime search and rescue, meteorological observation, and target detection. Fig. 1 illustrates an airborne radar tracking a maritime vessel target and the anti-jamming scenario, which includes noise suppression jamming and false target deception jamming. Due to weather, manpower, and material constraints, field experiments for this scenario are challenging. As shown in Fig. 2, a microwave anechoic chamber replicates the operational environment and electromagnetic signals, enabling high-precision simulation experiments for evaluating radar anti-jamming performance. However, each trial in the chamber still requires substantial measurement effort, making the experiment a genuinely expensive data source where one query corresponds to one physical measurement that takes hours.

The seven-dimensional input comprises the factors summarized in Table I. The three responses are summarized in Table II. The experiments start with an initial training set of 20 samples, followed by 50 adaptively selected samples, for a total budget of 70 samples. Each step selects the next query from a candidate pool of 100 points.

E. HPO-mixed

The expensive source is one complete training run of a LightGBM binary classifier on a stratified 50,000-row subset of the public Covertype dataset (581,012 rows, 54 features, 7 classes; the binary task is class 2 vs. rest), taking about 2.6 s per run. The search space consists of the eight LightGBM hyperparameters listed in Table III, which mixes quantity scales chosen to follow the conventions of the hyperparameter-optimization literature: the learning rate and the two regularization coefficients span four to eight orders of magnitude and are therefore log-scaled, whereas num_leaves, feature_fraction, and bagging_fraction are linear. Two design points deserve mention. First, num_leaves is scaled linearly rather than logarithmically so that the high-capacity regime (where the overfitting behavior of the categorical output concentrates) remains reachable by the sampler. Second, max_depth is a discrete-valued parameter in LightGBM (-1 means unconstrained); the continuous axis reserves its lowest 15% mass for the unconstrained setting and spreads the remaining mass uniformly over $[3, 16]$.

The product of the eight ranges yields a combinatorial configuration space far too large to enumerate ($\sim 10^{11}$ axis-product combinations), and, because each configuration requires a full training run, the source is treated as a black box: queries return only the measured responses, with no gradient or closed-form structure. To make the evaluation exact and reproducible, a precomputed truth table of 12,000 configurations is generated offline (each configuration evaluated with the random seed derived deterministically from its own hash), and this table serves as the black-box oracle for all arms: a query returns the recorded responses of the nearest table entry,

TABLE III

LIGHTGBM HYPERPARAMETER SEARCH SPACE OF HPO-MIXED. ALL EIGHT PARAMETERS ARE NORMALIZED TO $[0,1]^8$; THE MAPPING COLUMN STATES HOW A NORMALIZED VALUE $x \in [0,1]$ IS DECODED INTO THE NATIVE RANGE.

Dim.	Hyperparameter	Native range	Mapping
x_1	learning rate	$[10^{-4}, 1]$	10^{-4+4x_1} (log)
x_2	num_leaves	$[2, 256]$	$\mathrm{round}(2 + 254x_2)$ (linear)
x_3	min_data_in_leaf	$[2, 200]$	$\mathrm{round}(2 \cdot 100^{x_3})$ (log)
x_4	feature_fraction	$[0.3, 1.0]$	$0.3 + 0.7x_4$ (linear)
x_5	bagging_fraction	$[0.3, 1.0]$	$0.3 + 0.7x_5$ (linear)
x_6	λ_1 (ℓ_1 reg.)	$[10^{-8}, 10]$	10^{-8+9x_6} (log)
x_7	λ_2 (ℓ_2 reg.)	$[10^{-8}, 10]$	10^{-8+9x_7} (log)
x_8	max_depth	-1 or $[3, 16]$	$x_8 < 0.15 \to -1$; else 3–16

with no gradient or closed-form structure exposed. Training uses a fixed 500 boosting rounds without early stopping and one thread.

The continuous output y_1 is the validation accuracy on a held-out 20,000-row subset. The categorical output $y_2 = \mathbf{1}[\text{train acc} - \text{val acc} > \tau]$ with $\tau = 0.05$ labels overfitting. The experiments start with an initial training set of 20 configurations, followed by 50 sequential steps, for a total budget of 70 evaluations per arm and seed, over ten seeds with shared candidate pools of 2,000 configurations drawn from the truth table.

F. Chicago Crime

The source is the public Chicago crime dataset. The year 2021 (202,870 cleaned incidents) serves as the knowledge year and the year 2022 (233,049 cleaned incidents) is the target year that can only be queried within budget. The study area is a 1 km grid of 659 occupied cells over the city bounding box. Fig. 3 shows the spatial distribution of incidents in both years together with the dual-channel knowledge field constructed from the 2021 data.

One acquisition reveals both responses of a grid cell: the continuous response y_1 is the annual incident count (modeled on the $\log(1 + y_1)$ scale), and the categorical response $c = \mathbf{1}\{N_{2022} > Q_3\}$ is a hotspot label defined by the historical 2021 upper-quartile threshold ($Q_3 = 445$, positive rate 30%). Each seed holds out 132 cells for surrogate evaluation and shares an initial design of 30 cells plus 20 sequential acquisitions across all methods. After the budget is spent, every method's data pass through a common mixed-output surrogate, a predicted intensity field, and a multi-modal search with the Glowworm Swarm Optimization for four compact boxes (side lengths 1–15% of the domain, size-regularized), whose true quality is measured on all 2022 incidents.

II. KNOWLEDGE-FIELD CONSTRUCTION

This section details, for each case, how the two knowledge channels π_{hg} (hard-to-fit) and π_{bnd} (boundary) are constructed from the domain knowledge stated in the main text. Every field is analytic, constructed at zero evaluations of the source. The fused field is $\pi = \max(\pi_{\mathrm{hg}}, \pi_{\mathrm{bnd}})$ throughout, and every channel is truncated to $[\epsilon, 1]$ with $\epsilon = 0.01$. The piBO baselines

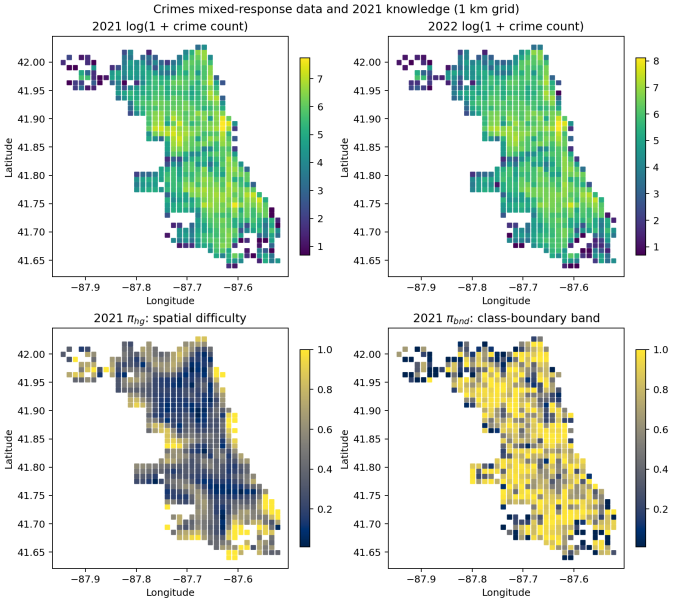


Fig. 3. Data and knowledge overview: incident distributions of the knowledge year (2021) and the target year (2022), and the dual-channel knowledge field constructed from 2021.

receive exactly the same fused field, statically and at full strength, so that any difference between KARMA and piBO is attributable to the region-level arbitrated injection alone.

A. General Construction Rules

The continuous cases draw on two primitives. The first is the *soft box*. For an axis-aligned box $B = [\mathrm{lo}, \mathrm{hi}]$ and a point $\mathbf{x} \in [0,1]^D$, let

$$d_{\mathrm{out}}(\mathbf{x}, B) = \left\| \mathbf{x} - \Pi_B(\mathbf{x}) \right\|_2, \tag{10}$$

where Π_B clamps each coordinate of $\mathbf{x}$ into its range in B, so the distance is zero inside the box. The soft-box field is

$$\pi_{\mathrm{box}}(\mathbf{x}) = \exp\Big[-\tfrac{1}{2} \big(d_{\mathrm{out}}(\mathbf{x}, B) / ew \big)^2 \Big], \tag{11}$$

i.e., one inside the box with a Gaussian decay of width ew outside. The second primitive is the *level-set band*. For a scalar statistic $s(\mathbf{x})$, a level c, and a core half-width w,

$$\pi_{\mathrm{band}}(\mathbf{x}) = \exp\Big[-\tfrac{1}{2} \big(\max\big(|s(\mathbf{x}) - c| - w,\, 0 \big) / ew \big)^2 \Big], \tag{12}$$

which is one on the core set $|s(\mathbf{x}) - c| \le w$ and decays as a Gaussian of width ew away from it. Unions of regions are formed by the maximum of the corresponding fields, and each final field is clipped to $[\epsilon, 1]$. The grid-based crime case uses the grid analogues of the same two primitives, given below.

B. Branin: Historical Experience

On Branin the knowledge plays the role of *historical experience from related tasks*. The hard-to-fit channel encodes the difficulty centers extracted from a Hessian-based curvature

analysis of an earlier system of the same type: three curvature-derived centers $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_3$ are closed into soft boxes of half-width 0.25 per axis,

$$\pi_{\mathrm{hg}}(\mathbf{x}) = \max_{j\in\{1,2,3\}} \pi_{\mathrm{box}}\big(\mathbf{x};\, [\mathbf{c}_j - 0.25,\, \mathbf{c}_j + 0.25],\, 0.06\big), \tag{13}$$

with a soft edge of width $ew = 0.06$. The boundary channel is a partially known limit-state band: the classification boundary is the level set $\log(y_{\mathrm{br}}) = 4$, and

$$\pi_{\mathrm{bnd}}(\mathbf{x}) = \pi_{\mathrm{band}}\big(\mathbf{x};\, \log y_{\mathrm{br}},\, 4,\, 0,\, 0.04\big) \tag{14}$$

peaks exactly on this level set and decays as a Gaussian of width 0.04 on both sides.

C. Ackley: Historical Experience, Wide Anchors

On Ackley the knowledge source is the same kind of *historical experience*, but the form differs to exercise the generality of the injection interface. The hard-to-fit channel uses wide difficulty anchors in place of the localized centers: representative anchor regions derived from the radial profile of the function form a union of three wide soft boxes,

$$\pi_{\mathrm{hg}}(\mathbf{x}) = \max_{j\in\{1,2,3\}} \pi_{\mathrm{box}}\big(\mathbf{x};\, [\mathbf{a}_j - 0.42,\, \mathbf{a}_j + 0.42],\, 0.12\big), \tag{15}$$

with anchor centers $\mathbf{a}_j$ and edge $ew = 0.12$. The boundary channel is a radial shell in place of the band: the limit state is the level function $g(\mathbf{x}) = y_1 y_2 + y_1 + y_2$ at level 35, and

$$\pi_{\mathrm{bnd}}(\mathbf{x}) = \pi_{\mathrm{band}}\big(\mathbf{x};\, g,\, 35,\, 0,\, \sigma_b\big), \tag{16}$$

where σ_b is the 10th percentile of $|g - 35|$ over a 2×10^5-sample calibration set, so the band width is learned from cheap function evaluations of the historical surrogate rather than fixed by hand.

D. Wingweight: Physics and Expert Intervals

On wingweight the knowledge combines *physics knowledge with expert experience intervals* from aircraft design. The hard-to-fit channel encodes three standard box-type regions over the design factors:

- thin-airfoil band: $t/c < 0.30$ (the factor $\kappa^{-0.3/-0.5}$ dominates the power-law curvature);
- light-load corner: $N_z < 0.35 \wedge W_{dg} < 0.35$ (the term $(N_z W_{dg})^{0.49/0.5}$ carries the second-largest curvature share);
- low-aspect-ratio band: $A < 0.25$,

each closed into a soft box with edge $ew = 0.08$ and fused by max. The boundary channel is the constraint-threshold band: the two design constraints $y_1 \le 250$ and $y_2 \le 50$ are rewritten as exact algebraic margins $e_1 = \log m_1 - \log(250 - S_w W_p)$ and $e_2 = \log m_2 - \log(50 + S_w W_p)$, where m_1, m_2 are the power-law main parts of y_1, y_2, and the field is the maximum of two Gaussian ridges over these margins,

$$\pi_{\mathrm{bnd}}(\mathbf{x}) = \max\left(e^{-\frac{1}{2}(e_1/0.025)^2},\, e^{-\frac{1}{2}(e_2/0.027)^2}\right), \tag{17}$$

so that the field peaks exactly on the union of the two constraint surfaces ($e_j = 0$) and decays with widths $\sigma_1 = 0.025$ and $\sigma_2 = 0.027$.

E. Ship: Electromagnetic Principles and Radar Practice

On ship the knowledge combines *electromagnetic principles with the experience of radar-domain engineers*. The hard-to-fit channel is a platform operating region derived from the physics: noise suppression is effective only at high SNR and the deceptive fake target is dangerous only when close, so the operating region $B_{\mathrm{phys}} = \{\mathrm{FTD\ SNR} \ge 0.6\} \times \{H_{\mathrm{dist}} \le 0.4\}$ is encoded as the soft box

$$\pi_{\mathrm{hg}}(\mathbf{x}) = \pi_{\mathrm{box}}(\mathbf{x}; B_{\mathrm{phys}},\, 0.06) \tag{18}$$

with $ew = 0.06$. The boundary channel is a sign criterion from engineering practice: the anti-jamming outcome is known to switch sign along the contrast function

$$s(\mathbf{x}) = \tfrac{1}{6}\,(x_4 + x_7 - x_2 - x_6 - x_1 - x_5), \tag{19}$$

and the field is the level-set band of s centered at the historical median $c = \mathrm{med}\,\{s\}$ over the historical pool with core half-width $w = 0.08$ and edge $ew = 0.06$.

F. HPO-mixed: Community Consensus and Overfitting Envelope

On HPO-mixed the knowledge encodes *community consensus on hyperparameter behavior*. The hard-to-fit channel is a soft band over the learning-rate sweet range: community consensus places good configurations at learning rates in $[0.03, 0.14]$ (normalized $x_1 \in [0.54, 0.76]$), so π_{hg} is the one-dimensional soft box on x_1 over $[0.54, 0.76]$ with $ew = 0.08$ and is constant along the remaining axes. The boundary channel is an overfitting-envelope band: the community knows that overfitting concentrates in the high-learning-rate, small-min-data corner, and the envelope is the outer shell of the box $O = \{x_1 \ge 0.70\} \times \{x_2 \ge 0.25\}$,

$$\pi_{\mathrm{bnd}}(\mathbf{x}) = \exp\Big[-\tfrac{1}{2}\big(d_{\mathrm{face}}(\mathbf{x}, O)/0.08\big)^2\Big], \tag{20}$$

$$d_{\mathrm{face}}(\mathbf{x}, O) = \min_{k\in\{1,2\}} \min\big(|x_k - \mathrm{lo}_k|,\, |x_k - \mathrm{hi}_k|\big), \tag{21}$$

i.e., one on the box faces and decaying with shell thickness 0.08 both inward and outward. Both channels are constructed from public documentation and a coarse pilot scan; the scan cost is two orders of magnitude below the evaluation budget.

G. Chicago Crime: Historical Intensity Structure

On the crime case the hard-to-fit channel is derived from the *2021 historical year only* and lives on the 1 km grid of occupied cells; it is the smoothed intensity-gradient field. Let c be the 2021 log-count grid (unoccupied cells masked out) and $\tilde{c}$ its mask-normalized Gaussian smoothing with $\sigma = 1$ cell, $\tilde{c} = G_\sigma(c \cdot \mathbb{1}) / G_\sigma(\mathbb{1})$; the field is the gradient magnitude normalized by its 95th percentile over the valid cells,

$$\pi_{\mathrm{hg}}(i) = \mathrm{clip}\Big(\|\nabla\tilde{c}\,(i)\|_2 \Big/ Q_{0.95}\big(\{\|\nabla\tilde{c}\|_2\}_{\mathrm{valid}}\big),\, \epsilon,\, 1\Big), \tag{22}$$

where the gradient uses the 1 km cell spacing; it captures the spatial transition zones that are historically difficult to fit. The boundary channel follows the same adjacency construction in both calibers and differs only in the year of the records that

are screened. With hotspot labels $y_j^{(2)} = \mathbf{1}[y_{1,j} > 445]$ and cell centers $\mathbf{c}_j$, the neighborhood hotspot-side probability of cell i is the Gaussian-weighted average

$$p_i = \frac{\sum_j \exp\big(-\|\mathbf{c}_i - \mathbf{c}_j\|^2/2\sigma^2\big)\, y_j^{(2)}}{\sum_j \exp\big(-\|\mathbf{c}_i - \mathbf{c}_j\|^2/2\sigma^2\big)}, \qquad \sigma = 1 \text{ cell}, \tag{23}$$

and a cell is a classification-boundary cell when a four-neighbor cell carries the opposite label; the field multiplies the hotspot-side probability by a Gaussian of the distance to the nearest boundary cell $\mathcal{B}$,

$$\pi_{\mathrm{bnd}}(i) = \exp\Big(-\operatorname{dist}(i,\mathcal{B})^2/2\sigma^2\Big)\, p_i, \tag{24}$$

so that the field peaks on the boundary cells that lean toward the hotspot side. In the fully deployable caliber the labels, the boundary set, and the distances are computed on the *2021* grid. In the display caliber of the main results the same construction is *distilled from a preliminary screening of the public 2022 records*: the labels and the boundary set are computed on the 2022 grid, so the field marks the high-crime side of the 2022 hotspot boundary and carries partial and coarse guidance rather than the labels themselves.

III. MSPSO-Based Region Identification

In the ship case, the region-guided arms (KARMA and its variants, together with the region-based no-knowledge baseline) do not score the whole candidate set directly. At every acquisition step, a multi-objective particle swarm optimizer (MSPSO) first searches the normalized design space $\Omega = [0,1]^d$ ($d = 7$) for a small set of axis-aligned regions in which the current MMRGP surrogate is hardest to fit, and the next query is then selected from the union of these regions. This section documents the algorithm; every value in Table IV is the exact configuration used in the experiments.

Step 1: Particle Encoding and Initialization

Candidate regions are encoded as axis-aligned hyper-rectangles in a starting-point-plus-width representation. For the i-th particle at iteration t, the position vector is

$$P_{i,t} = \big[p_{i1,t}^{(s)},\, w_{i1,t},\, p_{i2,t}^{(s)},\, w_{i2,t},\, \ldots,\, p_{id,t}^{(s)},\, w_{id,t}\big], \tag{25}$$

where $p_{id,t}^{(s)}$ and $w_{id,t}$ denote the starting coordinate and the width of the d-th dimension. The particle decodes to the hyper-rectangle $R_{i,t} = \prod_d \big[p_{id,t}^{(s)},\, p_{id,t}^{(s)} + w_{id,t}\big]$, which must satisfy the boundary constraints $l_d \le p_{id,t}^{(s)}$ and $p_{id,t}^{(s)} + w_{id,t} \le u_d$, together with the window proportion bounds $w_{\min} \le w_{id,t} \le w_{\max}$. The swarm size is $N = 50$. Initialization is size-first: the width of each dimension is drawn before the starting coordinate, and candidate particles overlapping already accepted regions are resampled. The particles, their personal bests, and their velocities are initialized by Latin hypercube sampling.

Step 2: Fitness Function Computation

For each candidate region $R_{i,t}$, $N_s = 50$ samples are drawn inside the region by Latin hypercube sampling, and the objective vector $F(R_{i,t}) = [\mathrm{IL}, G, E]$ is computed. All responses enter through the MMRGP predictions: the continuous responses through their predicted values and the categorical response through its latent positive-class probability; all values are min–max normalized over the sample set before aggregation.

The information landscape entropy (IL) captures regional ruggedness. Let $\{\mathbf{x}_i\}_{i=1}^n$ be the n samples in the region, q the number of responses, and $\tilde{f}_k(\mathbf{x})$ the normalized prediction of the k-th response. With tie tolerance $\epsilon = 0.05$, the preference between two samples is averaged over responses,

$$V_p(i,j) = \frac{1}{q}\sum_{k=1}^{q} \phi\big(\tilde{f}_k(\mathbf{x}_i), \tilde{f}_k(\mathbf{x}_j)\big), \tag{26}$$

$$\phi(a,b) = \begin{cases} 1, & a < b - \epsilon, \\ 0.5, & |a-b| \le \epsilon, \\ 0, & \text{otherwise}, \end{cases} \tag{27}$$

and the reference vector $V_r(i,j)$ encodes, in the same form, the ranking induced by the distance to the reference point $\mathbf{x}^*$, the sample whose normalized response vector is closest to the ideal point. The IL is then

$$\mathrm{IL} = \frac{2}{n(n-1)} \sum_{i<j} \big|V_p(i,j) - V_r(i,j)\big|. \tag{28}$$

The maximum gradient consistency (G) captures local fluctuation intensity: each sample is paired with its nearest neighbor $\nu(i)$ in the input space, and

$$G = \max_{1\le i\le n} \frac{s}{d_i + \epsilon} \cdot \frac{1}{q}\sum_{k=1}^{q} \big|\tilde{f}_k(\mathbf{x}_{\nu(i)}) - \tilde{f}_k(\mathbf{x}_i)\big|, \tag{29}$$

where d_i is the nearest-neighbor distance and s the input-space scale. The normalized mean entropy (E) quantifies categorical uncertainty: with $p_i^{(k)}$ the positive-class probability of the categorical response at $\mathbf{x}_i$ and $H(p) = -p\ln p - (1-p)\ln(1-p)$ the binary entropy,

$$E = \frac{1}{n\ln 2}\sum_{i=1}^{n} H\big(p_i^{(k)}\big). \tag{30}$$

A larger IL indicates a more rugged landscape, a larger G stronger local variation, and a larger E higher classification uncertainty.

Step 3: Archive and Leader Selection

After fitness evaluation, the external archive is updated with all current particles and their objective vectors, dominated solutions are removed, and, once the archive exceeds its capacity N, only the members with the largest crowding distances are retained, so that the archive traces the Pareto front of the three indicators. The global leader of each iteration is chosen by a binary tournament inside the current first front:

with probability 0.7 the member with the higher min–max-normalized mean quality score wins (convergence), and with probability 0.3 the member with the larger crowding distance wins (diversity).

Step 4: Particle Swarm Update

Velocity and position follow the standard PSO rules,

$$V_{i,t+1} = \omega V_{i,t} + c_1 r_1 \big(\tilde{P}_i - P_{i,t}\big) + c_2 r_2 \big(\bar{P}_t - P_{i,t}\big), \quad (31)$$

$$P_{i,t+1} = P_{i,t} + V_{i,t+1}, \quad (32)$$

with the velocity clamped component-wise to $\pm 20\%$ of the corresponding variable range and boundary/window feasibility corrections applied after each update. The factors are scheduled over the elapsed fraction $\rho_t \in [0, 1]$ of the current restart cycle,

$$\begin{aligned} \omega_t &= \omega_{\max} - (\omega_{\max} - \omega_{\min})\rho_t, \\ c_{1,t} &= c_{1,\max} - (c_{1,\max} - c_{1,\min})\rho_t, \\ c_{2,t} &= c_{2,\min} + (c_{2,\max} - c_{2,\min})\rho_t, \end{aligned} \quad (33)$$

with $\omega_{\max} = 0.9$, $\omega_{\min} = 0.4$, $c_{1,\max} = c_{2,\max} = 2.5$, and $c_{1,\min} = c_{2,\min} = 0.5$; the schedules are reset at every restart (Step 5).

Step 5: Adaptive Restart and Region Harvest

Let HV_t denote the hypervolume of the objective space dominated by the current archive, measured with respect to a reference point fixed at its first evaluation and computed by Monte Carlo sampling with 20,000 draws. A restart is triggered when the relative hypervolume improvement over the last recorded value stays below $\tau_{\mathrm{HV}} = 10^{-3}$ for p consecutive generations, where the patience is $p = \max(10, \mathrm{round}(T_R/5))$ and T_R is the per-region iteration budget (Table IV); the archive must contain at least five members before a restart is allowed, and three consecutive restarts that fail to produce a new region stop the restart mechanism. Upon restart, all particles, personal bests, leaders, and the schedule factors are reinitialized, while the archive is retained to preserve the search history. Each restart harvests one representative region: the non-dominated members that do not overlap already occupied regions are ranked by crowding distance, and one member is selected uniformly at random among the largest $k = 2M$ of them, where M is the number of objectives. The number of regions identified in a step is therefore an emergent outcome of the restart process rather than a fixed quota (about five regions per step on average in our runs under the no-knowledge configuration).

Step 6: Termination and Prior-Informed Variant

Each acquisition step runs one MSPSO call of at most $T = 200$ iterations, and the union of the harvested representative regions forms the candidate region set from which the next query is selected. In the knowledge-guided arms, the optimizer appends a fourth objective $\Phi(R)$, the annealed knowledge-consistency score defined in the main text, evaluated as the region mean of the fused knowledge field, so the search trades off indicator difficulty against knowledge alignment; the no-knowledge baseline uses the plain three-objective version above.

TABLE IV
PARAMETER SETTINGS OF THE MSPSO-BASED REGION IDENTIFICATION ALGORITHM (SHIP CASE). THE REGION BUDGET IS 14 FOR THE KNOWLEDGE-GUIDED ARMS AND 4 FOR THE NO-KNOWLEDGE BASELINE; THE RESTART-CYCLE BUDGET FOLLOWS THE SAME ORDER.

Parameter	Symbol	Value
Input dimension	d	7
Swarm size	N	50
Region evaluation samples	N_s	50 (LHS)
Tie tolerance	ϵ	0.05
Maximum iterations per call	T	200
Region budget	R_{roi}	14 / 4
Restart-cycle budget	T_R	$\mathrm{round}(T/R_{\mathrm{roi}})$ = 14 / 50
Restart patience	p	$\max(10, \mathrm{round}(T_R/5))$ = 10
HV improvement threshold	τ_{HV}	10^{-3}
Min. archive size for restart	–	5
Blank-restart fuse	–	3
Archive capacity	–	$N = 50$
Leader selection	–	0.7 quality / 0.3 crowding
Selection size	k	$2M$ ($M = 3$ or 4)
Window proportion bounds	$[w_{\min}, w_{\max}]$	$[0.35, 0.85]$
Velocity clamp	$v_{\max}$	$0.2\times$ variable range
Inertia weight	ω	$0.9 \to 0.4$
Cognitive factor	c_1	$2.5 \to 0.5$
Social factor	c_2	$0.5 \to 2.5$
HV Monte Carlo samples	–	20,000

IV. BASELINE METHODS

This section provides a brief introduction to the baseline methods used in the experimental comparison. All methods are adapted to the mixed-response setting by replacing their original single-output or homogeneous multi-response surrogate with the mixed-output Gaussian process surrogate described in the main paper. All methods share the same initial design, the same candidate pool, and the same budget per seed.

A. Maximin-LHS

Maximin Latin hypercube sampling is a space-filling design that does not consult the surrogate. A Latin hypercube design partitions each axis into equal-probability bins and places exactly one point per bin per dimension; the maximin criterion selects, among a set of candidate LHS designs, the one that maximizes the minimum pairwise distance. It serves as the no-feedback lower bound: any sequential method must beat it to justify its overhead.

B. MeanMSE

The mean predictive variance (MeanMSE) method uses the average predictive variance across responses as the acquisition criterion:

$$x_{\mathrm{next}} = \arg\max_{x \in \mathcal{X}} \frac{1}{m} \sum_{i=1}^{m} \sigma_i^2(x), \quad (34)$$

where $\sigma_i^2(x)$ denotes the predictive variance of the i-th continuous response at x. Compared with a per-response maximum, MeanMSE balances the information gain over all continuous responses. Nevertheless, because every response is

weighted equally, the criterion may be dominated by high-variance outputs, and it does not account for categorical-response boundaries.

C. IMSE

The integrated mean squared error (IMSE) method selects the sample that minimizes the integrated predictive variance over the entire input space. Let $p(x)$ be a probability density over $\mathcal{X}$ (typically uniform). After augmenting the training set with candidate x, the integrated variance is

$$x_{\text{next}} = \arg\min_{x \in \mathcal{X}} \int_{\mathcal{X}} \sum_{i=1}^{m} \sigma_i^2(x' \mid D_n \cup \{(x, y)\})\, p(x')\, dx', \tag{35}$$

where $\sigma_i^2(x' \mid D_n \cup \{(x, y)\})$ is the predictive variance at x' after adding x. IMSE is a global criterion. It tends to spread samples evenly and can waste evaluations in flat regions where the surrogate is already accurate.

D. PALC

The partitioned active learning for heterogeneous systems (PALC) method targets spatially heterogeneous response surfaces. Rather than relying on a single global surrogate, PALC builds a partitioned Gaussian process (PGP) by splitting the design space $\mathcal{X}$ into M disjoint subregions $\{\Omega^{(1)}, \ldots, \Omega^{(M)}\}$ and fitting an independent local GP in each region. The resulting block-diagonal covariance structure both shields each region from distortion caused by heterogeneous neighboring data and reduces the cubic inversion cost.

Global searching. At each iteration PALC first selects the subregion with the largest integrated predictive uncertainty. Let $\sigma^{2(m)}(s)$ denote the predictive variance of the m-th local GP at s. The target subregion is

$$m^* = \arg\max_{m=1,\ldots,M} \int_{\Omega^{(m)}} \sigma^{2(m)}(s)\, p(s)\, ds, \tag{36}$$

where $p(s)$ is a probability density over $\Omega^{(m)}$ (typically uniform). This macro step narrows the candidate space from the full design space to the single most informative region.

Local searching. Once m^* is fixed, the next sample is chosen inside $\Omega^{(m^*)}$ by minimizing the partitioned integrated mean squared error (PIMSE):

$$x_{\text{next}} = \arg\min_{x \in \Omega^{(m^*)}} \int_{\Omega^{(m^*)}} \sigma^{2(m^*)}(s \mid D_n \cup \{(x, y)\})\, p(s)\, ds. \tag{37}$$

In our mixed-response adaptation, each local GP is replaced by the mixed-output surrogate. The regional uncertainty aggregates the predictive variances of all continuous responses. The categorical boundary is handled as an additional heterogeneity indicator that guides the partitioning of the input space.

E. AK-MCS-IRS

The adaptive Kriging Monte Carlo simulation with important region sampling (AK-MCS-IRS) method combines a Kriging surrogate with Monte Carlo simulation and importance sampling for reliability analysis. The core idea is to construct a learning function that identifies points closest to the limit state surface. A commonly used learning function is the U-function:

$$U(x) = \frac{|\hat{g}(x)|}{\sigma_g(x)}, \tag{38}$$

where $\hat{g}(x)$ and $\sigma_g(x)$ are the predictive mean and standard deviation, respectively. The next sample is chosen as the point with the smallest U-value, i.e., the point whose sign is most uncertain. Once the surrogate is sufficiently accurate, the failure probability is estimated by importance sampling.

F. KeyRegion

The KeyRegion method first identifies a set of important regions from the surrogate (based on combined uncertainty, sample-distance, and response-magnitude criteria), then selects the next sample inside the region with the highest score. It controls for region structure without injected knowledge.

G. Voronoi

The Voronoi baseline partitions the design space into Voronoi cells around the current samples and selects the next sample in the cell with the largest combined metric (predictive uncertainty times cell volume), enforcing geometric coverage of the space. It controls for geometrically driven diversity.

H. HHs

The hierarchical hypersphere (HHs) method organizes the input space as a hierarchy of hyperspheres around labeled samples and selects the next sample on the hypersphere shell that maximizes a mixed-response score that combines the continuous prediction margin and the categorical latent-variable boundary proximity. It controls for shell-based local search with a mixed-response score.

I. piBO-Global and piBO-ROI

The two piBO variants implement fixed point-level prior weighting. Both receive the same fused knowledge field π as KARMA, statically and at full strength. piBO-global scores every candidate in the global pool as $\text{score}(x) = M_{\text{data}}(x) \cdot \pi(x)^{\beta/n}$, where M_{data} is the unchanged data-driven metric and β/n is the piBO decay schedule with $\beta = (\text{total budget} - \text{initial budget})/10$. piBO-ROI first identifies a data-driven ROI (identical to the no-knowledge variant) and then applies the same weighted scoring to the candidates inside it. These two baselines control for knowledge injected with fixed pointwise weighting instead of region-level arbitration.

J. QBC-Global

The query-by-committee baseline maintains a committee of $K = 8$ mixed-output surrogates, each refitted on an 80% class-stratified subsample of the current training set. The score of a candidate is the sum of the z-normalized classification disagreement (Jensen–Shannon divergence of the committee's class posteriors) and the z-normalized cross-committee standard deviation of the continuous predictions. The candidate with the highest score in the pool is selected. It controls for whether the gain of KARMA comes from active selection as such.

TABLE V
HPO-MIXED, TRUSTED BEST-FOUND (TBF) OF THE COMPARED METHODS, TEN SEEDS.

#	Entity	tBF-final	std
1	KARMA-EI (ours)	0.7804	0.0118
2	QBC-global	0.7789	0.0089
3	piBO-ROI	0.7786	0.0089
4	AK-MCS-IRS	0.7784	0.0029
5	KeyRegion	0.7779	0.0093
6	IMSE	0.7779	0.0091
7	BO-EI (acquisition)	0.7776	0.0038
8	MeanMSE	0.7762	0.0106
9	PAL-ROI-SUR	0.7759	0.0040
10	HHs	0.7755	0.0049
11	piBO-global	0.7753	0.0046
12	Voronoi	0.7744	0.0058
13	MaximinLHS	0.7733	0.0059
Truth-table trusted optimum		0.8022	
Top-1% validation line (99% overfitted)		0.7927	

TABLE VI
CHICAGO CRIME CASE, COMPLETE FIFTEEN-METHOD TABLE, TEN SEEDS, B=50. REGION METRICS ARE COMPUTED ON ALL 2022 INCIDENTS. BEST IN BOLD.

Method	RP ↑	CRP ↑	DL ↑	TopN ↑
Maximin-LHS	0.2500	0.3250	0.9890	0.2939
MeanMSE	0.3000	0.5000	1.0326	0.3515
IMSE	0.3000	0.3500	1.0333	0.3362
PAL-ROI	0.3750	0.4250	1.0618	0.3496
AK-MCS-IRS	0.3750	0.4250	1.0149	0.3341
piBO-global	0.3750	0.4750	1.0535	0.3514
piBO-ROI	0.3500	0.3750	0.9900	0.3268
KeyRegion	0.4250	0.4250	1.0809	0.3621
Voronoi	**0.4750**	0.5750	1.0769	0.3664
HHs	0.3750	0.4750	1.0625	0.3603
QBC-global	0.3250	0.4000	1.0562	0.3447
ROI-D (no knowledge)	0.4000	0.5250	1.0760	0.3631
KARMA-2021 (deployable)	0.4250	0.5500	1.0615	0.3490
KARMA-HG (single channel)	0.4250	0.5500	1.0496	0.3536
KARMA(display caliber)	**0.4750**	**0.6500**	**1.1232**	**0.3842**

V. EXTENDED HPO-MIXED COMPARISON

Table V reports the trusted best-found of the compared methods. The reference lines are the trusted optimum of the 12,000-configuration truth table (0.8022) and its top-1% validation-accuracy threshold (0.7927, of which 99% of configurations are overfitted).

The plain best-found (BF) curves of Fig. 4 complement the trusted-caliber curves of the main text and support two observations. First, once the overfitting restriction is removed, the expected-improvement criterion shows its true strength: BO-EI, which operates on the global pool with the classical EI acquisition, leads the comparison for most of the budget and converges to 0.7978, well above the remaining families. Second, KARMA-EI stays level with BO-EI throughout the search while additionally keeping every selection within the trustworthy region, and in the final stage it overtakes BO-EI and delivers the best configuration of the whole comparison (0.7982). In other words, under the unrestricted caliber the classical EI criterion is a strong frontier, and the knowledge-guided region variant matches that frontier pointwise while remaining eligible for the trusted accounting of the main text.

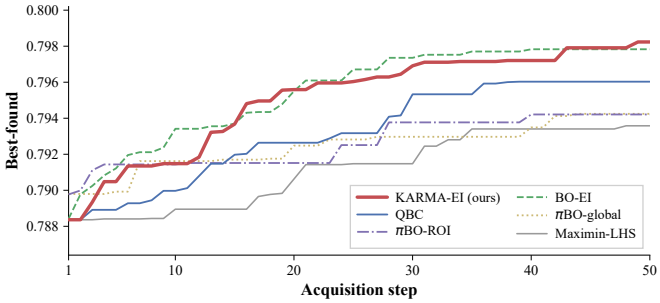


Fig. 4. Plain best-found (BF) convergence on HPO-mixed over ten seeds, without the overfitting restriction. Under this caliber the expected-improvement criterion operates at its full strength: BO-EI leads for most of the budget, and KARMA-EI runs level with it throughout the search and overtakes it in the final stage to deliver the best configuration overall.

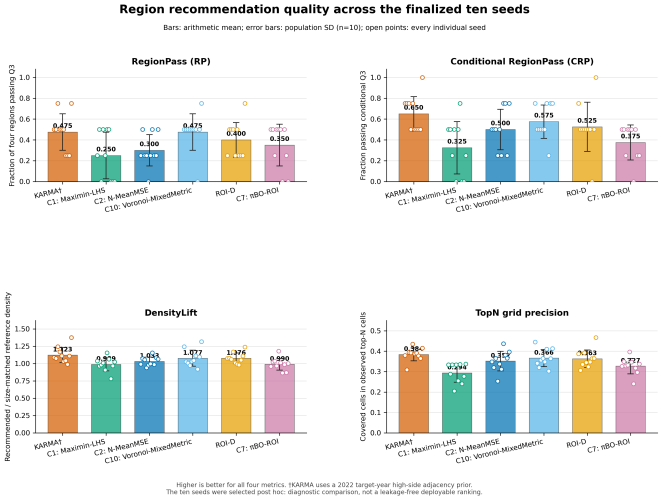


Fig. 5. Region metrics across the ten seeds for six representative arms; bars show ten-seed means, error bars show the population SD, and open points show individual seeds.

VI. CHICAGO CRIME CASE: FULL TABLE AND DIAGNOSTICS

Table VI is the complete fifteen-method table corresponding to the compact twelve-method comparison in the main text. Fig. 5 shows the distribution of the region metrics across the ten seeds, and Fig. 6 extends the spatial view of the main text to all twelve methods: for every method, the left panel shows the mixed-output surrogate prediction trained on its own 50-cell design together with the acquired samples and the four recommended regions, and the right panel shows the same regions over the 2022 observed density.

a) Caliber statement: The finalized display configuration derives its boundary channel from target-year labels, which we state openly in the main text as an oracle-style upper bound on aligned knowledge; the fully deployable configuration with 2021-only knowledge (KARMA-2021) is included in the same table. The ten finalized seeds are a post-hoc subset of thirteen completed seeds, selected by a pre-declared rule on the oracle reference; the thirteen-seed pool is retained in the repository

for sensitivity checks.

VII. Reproducibility Statement

Every experiment of the paper is reproducible from the released repository, which contains: per-case one-stop data archives with the finalized seed sets and all arm trajectories; the exact drivers, samplers, and knowledge-field constructors; the plotting and table-generation scripts used for the paper; and the per-case experiment records documenting protocol decisions, seed provenance, and known caveats. The environment is MATLAB R2022a with GPML for the mixed-output surrogate and Python 3.9 for the public-data pipeline. The repository is provided through an anonymized open URL, per the ICDE supplemental-material policy.

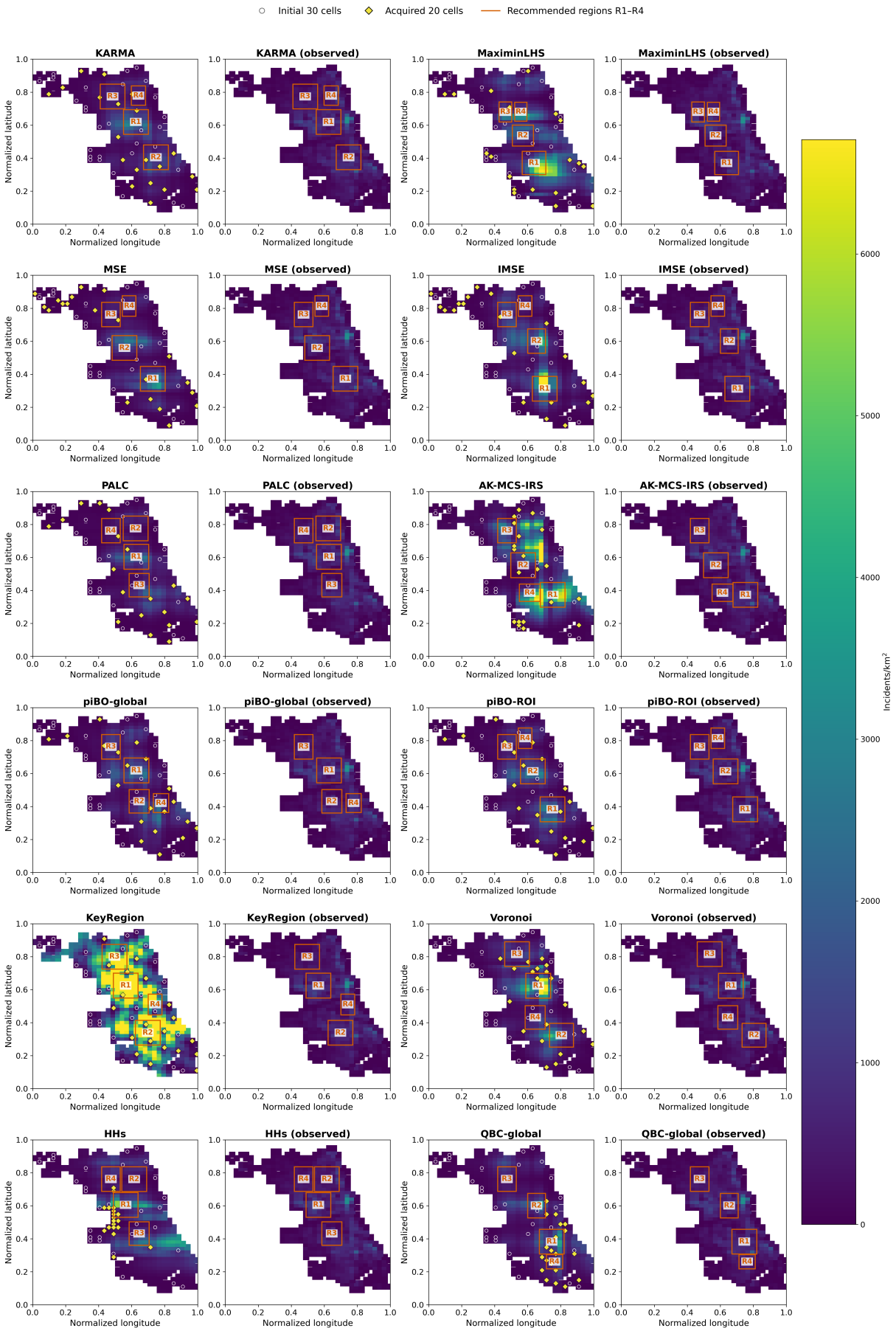


Fig. 6. Recommended regions of all twelve methods of Table V on a representative seed, overlaid on the 2022 incident density. For each method, the left panel shows the surrogate prediction with the 50-cell design and the four recommended regions, and the right panel shows the same regions over the 2022 observed density.